
FactoryBench: Benchmarking Machine Understanding via Question-Answering

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Abstract

Time-series models are widely used in industrial monitoring tasks such as forecasting, anomaly detection, and signal analysis. While highly effective for these objectives, they are often opaque and limited in their ability to provide structured reasoning for engineering decision-making. Large language models (LLMs), in contrast, can generate coherent, context-aware explanations and support multi-step reasoning. However, general-purpose LLMs still struggle when applied directly to dense multivariate sensor streams and machine-specific diagnostics without adaptation or external tools. A promising direction is therefore to use LLM agents augmented with specialized numerical and signal-processing tools. This raises a central question: how can we verify that these systems truly understand machine behavior rather than only generating plausible text?

We introduce **FactoryBench**, a benchmark for evaluating LLM agents on machine understanding over industrial time-series data. Our contributions are threefold. First, we propose a scalable framework for generating machine-understanding question-answering tasks, built around 80 structured question templates spanning diverse reasoning scenarios. Second, we present **FactoryWave**, a dense multivariate dataset generated from one-armed robotic systems, including both collaborative and industrial manufacturing robots. Third, we construct FactoryBench as a large-scale Q&A dataset for time-series understanding in LLM agents, grounded in FactoryWave as well as open-source robotics datasets and simulations. Together, these components provide a rigorous testbed for evaluating reasoning, causal understanding, and decision support over real and simulated industrial signals.

1 Introduction

Modern industrial systems generate large volumes of multivariate time-series data from sensors, actuators, and controllers. In robotic manufacturing settings, these signals encode joint states, torques, forces, velocities, contact events, task phases, and fault indicators. Extracting actionable knowledge from such data is central to monitoring, diagnostics, anomaly detection, and decision support.

Traditional time-series models excel at narrow tasks such as prediction, classification, and anomaly detection. However, these models are often specialized and difficult to interpret. They typically output labels or numeric predictions without explicit reasoning or higher-level explanations and cannot do much more outside of the specific task they are trained on, which limits their direct utility for automated engineering decision-making.

Large language models, on the other hand, exhibit strong general reasoning abilities and can produce structured explanations in technical contexts Wei et al. [2022], Yao et al. [2023b]. They can integrate textual information, follow logical chains, and generate interpretable outputs. Yet, when applied directly to dense numerical time series, general LLMs still underperform without adaptation. At the same time, transformer-based architectures have become increasingly effective for time-series forecasting and representation learning Vaswani et al. [2017], Zhou et al. [2021], Wu et al. [2021], Zhou et al. [2022], Nie et al. [2023], Woo et al. [2024], Das et al. [2024].

A promising paradigm is to build LLM agents equipped with specialized tools, including time-series models and signal-processing modules Schick et al. [2023], Yao et al. [2023a], Qin et al. [2023]. These agents can combine language reasoning with domain-specific computation. Nevertheless, evaluating whether such systems truly understand machine behavior remains challenging. Existing benchmarks mostly target textual reasoning, code generation, or generic multimodal understanding Hendrycks et al. [2021], Srivastava et al. [2023], Yue et al. [2024], leaving a gap in rigorous evaluation for machine-centered, time-series reasoning.

To address this gap, we propose FactoryBench, a benchmark designed to evaluate machine understanding in LLM agents. FactoryBench is grounded in FactoryWave, a dense multivariate time-series dataset collected from one-armed robotic systems, including collaborative and industrial manufacturing robots. The dataset captures rich dynamic behavior under diverse operational conditions.

We also introduce a scalable question-generation framework composed of 80 structured templates spanning a wide range of reasoning types and applicable to general machine data flows. These templates enable systematic construction of question-answering tasks that probe state understanding, intervention reasoning, counterfactual analysis, and decision-making in machine contexts. Using this framework and the FactoryWave dataset, we build FactoryBench, a large-scale Q&A dataset specifically designed to evaluate time-series understanding and engineering reasoning in LLM agents.

By bridging the gap between general language reasoning and specialized time-series modeling, FactoryBench provides a principled foundation for studying machine-centered intelligence in industrial environments.

2 Related work

2.1 LLM agents and tool use

Recent advances in LLM reasoning include prompting and deliberative strategies (e.g., Chain-of-Thought and Tree-of-Thought) that improve multi-step performance Wei et al. [2022], Yao et al. [2023b], alongside tool-augmented paradigms such as Toolformer and ReAct that enable grounded computation through external modules Schick et al. [2023], Yao et al. [2023a], Qin et al. [2023]. These ideas motivate industrial agent design where symbolic reasoning must be combined with numerical workflows.

2.2 Time-series learning for industrial signals

Time-series modeling has advanced through transformer-based architectures (Informer, Autoformer, FEDformer, PatchTST) Zhou et al. [2021], Wu et al. [2021], Zhou et al. [2022], Nie et al. [2023], strong alternative baselines such as TFT and N-BEATS Lim et al. [2021], Oreshkin et al. [2020], and critical comparisons against simpler linear models Zeng et al. [2023]. Foundation-model directions (e.g., Chronos) further explore transfer and zero/few-shot adaptation Woo et al. [2024], Das et al. [2024]. In parallel, anomaly and reliability monitoring literature includes OmniAnomaly, USAD, and Deep SVDD Su et al. [2019], Audibert et al. [2020], Ruff et al. [2018], with surveys and benchmarks highlighting persistent gaps in interpretability and root-cause actionability Chalapathy and Chawla [2019], Lavin and Ahmad [2015].

2.3 Causality and benchmark gaps

Causal frameworks provide formal tools for interventions and counterfactuals Pearl [2009], Peters et al. [2017], Rubin [1974], while temporal methods such as Granger causality and modern nonlinear causal discovery support directional reasoning in time series Granger [1969], Runge et al. [2019]. Existing broad benchmarks (MMLU, BIG-bench, MMMU) Hendrycks et al. [2021], Srivastava et al. [2023],

Table 1: Comparison of FactoryBench with existing benchmarks.

Benchmark	Machine/Robot	Multivariate TS	Size	Counterfactual	Ranking	Numerical	Free-Form	Novel Dense TS
TimeSeriesExam	×	×	~700	×	✓	×	×	×
ChatTS	×	✓	~134k	×	✓	✓	×	×
EngineMT-QA	✓	✓	~110k	×	✓	✓	✓	×
TSAQA	×	×	~210k	×	✓	×	×	×
Time-MQA	×	✓	~200k	×	✓	✓	✓	×
MTBench	×	✓	?	×	✓	✓	✓	×
QuAnTS	×	✓	~150k	×	✓	✓	✓	×
FactoryBench (Ours)	✓	✓	TBD	✓	✓	✓	✓	✓

Yue et al. [2024] and classical time-series resources Laptev et al. [2015], Dau et al. [2019], Wen et al. [2022] do not directly evaluate machine-centered reasoning over industrial telemetry. FactoryBench targets this gap by jointly testing temporal interpretation, causal reasoning, and engineering decision support.

3 FactoryBench framework

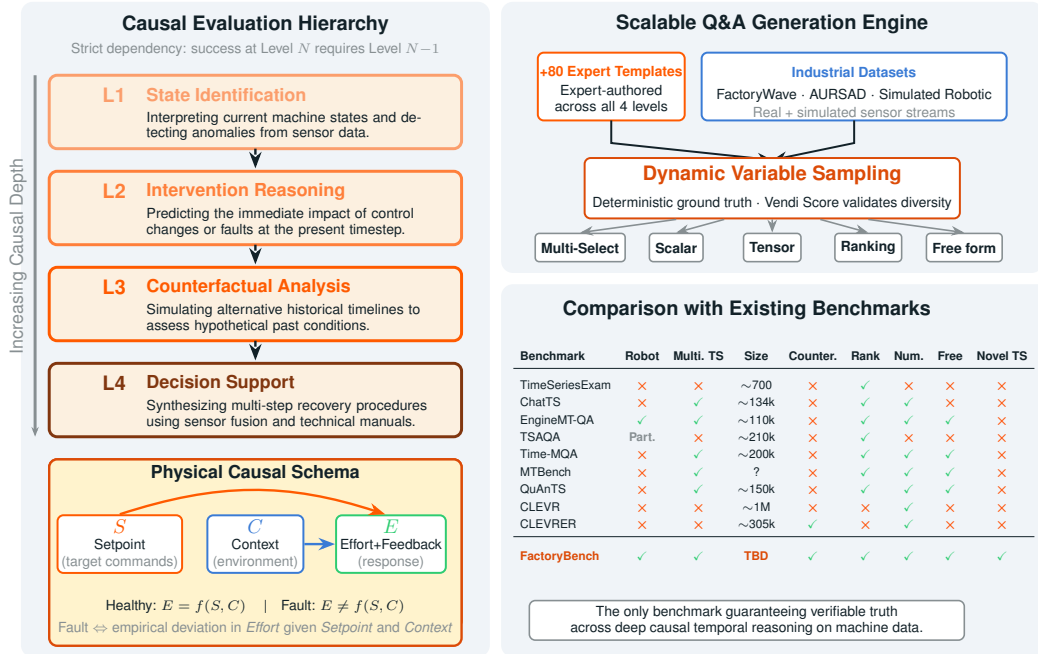


Figure 1: Overview of the FactoryBench framework. The left panel shows the four-level causal evaluation hierarchy and the physical causal schema. The right panel details the scalable Q&A generation engine and compares FactoryBench against existing benchmarks.

3.1 Four levels of machine understanding

To systematically evaluate machine understanding, we organize question-answering tasks according to a four-tier hierarchy, each probing distinct reasoning capabilities:

Table 2: Four levels of machine understanding in FactoryBench.

Level	Task	Example Question	Ground Source	Truth	Commercial Value
1	State	“What is the position tracking error of joint 3 at $T = 200\text{ms}$?”	Sensor data + KG		Fleet monitoring
2	Intervention	“If force in joint 3 increases <i>now</i> to X , what happens?”	Simulation (present state)		Diagnostic intervention
3	Counterfactual	“If force had increased to X at $t = 20\text{ms}$, what would have happened?”	Simulation (past state)	(past state)	Root cause / Capacity planning
4	Decision	“Robot stopped with error C203A. What to do?”	Manual + sensor fusion		Expert-free recovery

Level 1: State. This tier assesses the agent’s ability to read off the instantaneous or near-instantaneous state of the machine directly from its raw time-series streams. In our current implementation, Level 1 spans ten topic families covering universal physical quantities: (i) joint position displacement between two timestamps, (ii) a friction proxy defined as $|\text{effort_current}|/|\text{feedback_speed}|$, (iii) end-effector acceleration from vibration sensors, (iv) external force detection and magnitude, (v) joint jerk computed via central-difference approximation, (vi) torque magnitude derived from wrench data, (vii) position tracking error between commanded setpoint and measured feedback, (viii) joint speed, (ix) motor current draw, and (x) free-form natural language description of signal behavior over a window. Each topic is instantiated across multiple answer formats (True/False, multiple choice, open-ended numerical, and where meaningful, ranking and free-form), yielding a diverse question pool per underlying physical quantity. Two topic families additionally exploit the Machine Knowledge Graph stored in `data/labelling/machines.json`, which encodes per-joint rated speed and continuous current limits, to probe whether the agent can reason about sensor readings relative to documented operational envelopes rather than against arbitrary thresholds. All templates are designed to be robot-agnostic: they operate on a normalized signal schema (e.g., `feedback_pos_{i}`, `effort_current_{i}`) with parametric thresholds, so the same logic transfers to any manipulator once its streams are mapped into the common format. Level 1 generation is integrated directly with the Hugging Face Hub and consumes raw Parquet files from the `Forgis/FactoryNet_Dataset` repository, which are then split into monotonic segments to guarantee time-series consistency before templates are evaluated. Free-form signal descriptions are scored via LLM-as-judge, while all remaining formats admit deterministic ground truth computed directly from the labelled episode data.

Level 2: Intervention. At this level, the agent must reason about the consequences of interventions or events occurring at the present timestep that might perturb the distribution of machine states. Tasks include predicting the immediate impact of control actions, diagnosing faults as they arise, and understanding causal relationships in real time.

Level 3: Counterfactual. This tier evaluates the agent’s ability to reason about hypothetical scenarios, such as the effect of an event or intervention at a previous timestep. Questions require the agent to simulate alternative histories and assess how outcomes would differ under counterfactual conditions, while still considering the history they know.

Level 4: Decision Making. The highest tier encompasses complex decision-making tasks, where the agent must generate a sequence of actions or recommendations based on the time series and a prompt. This includes troubleshooting, optimization, and planning, requiring integration of state interpretation, causal reasoning, and goal-directed synthesis.

By structuring Q&A tasks along these four levels, FactoryBench enables rigorous and granular assessment of machine understanding, from basic state recognition to advanced decision support.

Design principle: Each level builds on previous. Failure at Level N implies failure at Level $N + 1$.

3.2 Answer formats

FactoryBench uses three answer formats, chosen to balance evaluation rigor with scalability. Each format supports deterministic scoring (with the exception of free form), has been chosen in order to make questions very easily checkable, but also hard to answer in the absence of correct reasoning.

Table 3: Open-source datasets incorporated into FactoryBench.

Dataset	Robot	Size	Frequency	Task	# Anomaly Types	SCE Compliance
AURSAD	UR3e	?	100 Hz	Screwing	4	✓
voraus-AD	UR5	?	100/500 Hz	Pick and Place	12	✓

Multi-Select True/False. The model is presented with four independent statements (A–D) about the outcome of an event or intervention, and must classify each as True or False. This format tests whether the model understands the causal consequences of a perturbation across multiple dimensions simultaneously — safety state, tracking error, current draw — without the answer leaking from any single correct choice. Evaluation gives a score of 1 for exact match, 0.5 for 1 mistake and 0 otherwise (random guesser is expected to get 0).

Ranking. The model is given four time-series segments or outcomes (A–D) and must order them according to a specified criterion (e.g., severity of deviation, magnitude of a signal). The answer is a permutation string (e.g., DABC). This format probes the model’s ability to perform relative quantitative reasoning rather than threshold-based classification. Evaluation uses exact-match rate and Kendall’s τ rank correlation against the ground-truth ordering.

Tensor Prediction. The model must predict at least one specific scalar value—such as the expected sensor reading following an intervention—with no multiple-choice scaffolding. The answer is a list of floating-point numbers (possibly one). This format directly tests quantitative extrapolation from time-series context. Evaluation is done on each scalar separately, and uses mean absolute percentage error (MAPE) and a threshold-based accuracy metric (prediction within $\pm k\%$ of ground truth), with partial points given to all correct scalars given.

Free-Form. The model produces an open-ended natural language response such as a diagnosis, recommended action sequence, or causal explanation. Because no single deterministic answer exists, evaluation is performed via an LLM-as-judge voting protocol: three independent frontier LLMs each compare the model’s response against the ground-truth reference and cast a verdict of **Wrong** (0), **Neutral** (0.5), or **Correct** (1). The final score is the majority vote across the three judges, with ties broken by averaging.

3.3 Time series data and causal schema (SCE)

To ensure that the dataset structure itself encodes causality, FactoryBench organizes all time-series signals into three causal groups, answering the core question: “*What was the machine told to do, and what did it actually do?*”

- **Setpoint:** The controller’s command—target position, velocity, and acceleration per axis.
- **Context:** Physical conditions affecting behavior—payload, temperature, material properties. Split into *static* (episode metadata) and *dynamic* (time-series columns).
- **Effort + Feedback:** The machine’s response—motor current (effort), actual position (feedback), vibration, and acoustic emission.

This structure enables a universal fault definition: under healthy operation, Effort is a lawful function of both Setpoint and Context, i.e., $E = f(S, C)$. Faults manifest as empirical deviations from this relationship. Faults manifest as deviations in the $f(\text{Setpoint})$ vs Effort relationship. The dataset provides the paired signals that make this comparison possible.

This data is sourced from:

- **FactoryWave:** A custom dataset generated from one-arm robotic platforms executing canonical industrial tasks (e.g., pick-and-place) under varied conditions and systematically injected anomalies.
- **Open Source Datasets:** Adapted datasets such as Aursad and Vorausad, offering diverse industrial scenarios and preprocessed to conform to the unified episode structure.

- **Simulations:** Synthetic time-series data generated from simulated robotic systems to enable controlled experimentation, ablation studies, and validation across both physical and virtual domains.

4 Reliable Q&A generation (key contribution)

4.1 At scale generation via extensive labelling

Scalable ground-truth generation is the central challenge of any Q&A benchmark grounded in raw sensor data. FactoryBench addresses this by coupling a structured labelling ontology with a context-free grammar (CFG)-style template system, designed by PhD-level experts in robotics. Rather than annotating individual questions by hand, each template is parameterized: concrete values are filled at generation time from the episode data and its associated labels. The context time series used to fill the variables of the question is sampled uniformly by datasource (sim vs open source vs FactoryWave), dataset (if looking at open source), experiment, difficulty, length and then placement in that order. This yields a combinatorial expansion from a small set of carefully designed templates into a large, diverse question pool.

Variable sampling. Each question template contains multiple variable slots that are filled at generation time via variable-specific sampling distributions. Continuous variables—such as signal names, timestamps, prediction horizons, and numerical thresholds—are sampled directly from the episode data or drawn from predefined distributions. Discrete variables—such as tasks, anomaly types, and root causes—are sampled from curated vocabularies. These vocabularies were initially aggregated from the open-source datasets used in FactoryBench, then substantially expanded by PhD-level robotics experts to cover a broader range of operationally realistic scenarios, while remaining fully reproducible for FactoryWave episodes. This separation between template structure and sampled content is what allows a small number of hand-authored templates to generate a large and semantically diverse question pool.

Template design. Each of the 80+ question templates was manually authored to probe one specific reasoning capability at the appropriate level of the hierarchy, while remaining general enough to admit a wide range of concrete instantiations. Templates are parameterized over episode segments, signal names, event descriptions, timestamps, and predicted values. Answer options for multi-select questions are drawn from a shared pool of verifiable statements, each paired with a rule that can be evaluated deterministically against the time series, given the densely labelled data. This design ensures that ground truth is never imputed or inferred—it is computed directly from labeled episode data—making the benchmark both reliable and fully reproducible.

4.2 Density of FactoryWave

4.3 Adapting labelling to open datasets and simulations

4.4 Dataset diversity & quality assurance

A critical risk in template-generated benchmarks is a lack of semantic diversity, leading models to memorize structural patterns rather than perform true reasoning. To ensure our dataset evaluates robust machine understanding, we utilize the **Vendi Score** on the embeddings of our Q&A pairs to measure and maximize effective population diversity.

We iteratively evaluate dataset diversity across three primary axes during generation:

- **Low Diversity (Parameter Variation):** Varying only parameters like time windows and joint indices yields a low Vendi Score, confirming that simple parameter randomization is insufficient.
- **Moderate Diversity (Format Variation):** Semantically identical questions expressed across different formats (Open-ended, Multiple Choice, True/False) measurably increase the effective diversity.
- **High Diversity (Template & Type Variation):** Introducing mixed reasoning templates across levels (e.g., kinematic comparison, derivative estimation like friction/acceleration,

Table 4: Expected performance across levels (accuracy %).

Level	Random	Current LLMs	Target (w/ prior)	Human Expert
1	50%	85–95%	98%	99%
2	20%	60–75%	85%	95%
3	10%	35–50%	70%	90%
4	5%	15–30%	45%	80%

anomaly detection) drives the highest Vendi Score. By enforcing high Vendi Scores across our generated subsets, we ensure the benchmark tests versatile analytical capabilities.

4.5 Pipeline overview

5 Experiments

5.1 Models to evaluate

Frontier LLMs:

- GPT-4o, GPT-4-Turbo
- Gemini-2.5-Pro, Gemini-2.5-Flash
- Claude-3.5-Sonnet, Claude-3-Opus

Specialized Methods:

- Time-series encoders + LLM (Chronos, Moirai)
- Multimodal industrial models (FD-LLM)
- RAG with manual retrieval

Baselines:

- Random
- Rule-based (threshold detection)
- Human expert (ceiling)

5.2 Experimental configurations

Beyond out-of-the-box evaluation, we investigate how different interventions improve performance:

- **Finetuning:** We optionally finetune open-source models on the FactoryBench Q&A dataset to establish the limit of specialized training versus general reasoning.
- **Tool-Augmented Agents:** The best-performing foundational models are granted access to external time-series prediction and anomaly detection modules, testing the synergy between LLM reasoning and domain-specific computation.

5.3 Results (expected)

Hypothesis: Semantic priors (ManualsGraph) will show largest gains at Level 4.

6 Discussion

6.1 Why this benchmark matters

For researchers: First rigorous evaluation of machine understanding across reasoning levels.

For practitioners: Quantifies which AI capabilities are ready for deployment as AI engineers.

For the field: Defines “machine understanding” operationally via Q&A.

6.2 Limitations

- Single machine family (by design, for depth)
- Simulation-to-real gap in synthetic episodes
- Reliance on LLMs-as-Judge for evaluation of Free-Form questions

6.3 Relation to FactoryNet

This benchmark evaluates on a **subset** of FactoryNet. The full dataset (50k+ episodes, 200k+ Q&A) will be released separately with a focus on *training* industrial world models, not just evaluation.

7 Conclusion

FactoryBench introduces a systematic approach to evaluating machine understanding via Q&A. By focusing deeply on one machine family and providing reliable answer generation at scale, we enable rigorous comparison of methods across four levels of reasoning—from basic state identification to procedure generation grounded in technical manuals.

Our physical validation protocol closes the loop from benchmark to reality, addressing the fundamental question: can AI systems that excel at industrial Q&A actually help in the real world?

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A Q&A pair structure

Level 2 Example (Intervention):

```
{
  "id": "4fab9ab-e476-48ff-812d-8fc33bb63ac3",
  "level": 2,
  "template_id": 2,
  "template_type": "intervention_outcome",
  "question": "If a continual increase of feedback_tcp_speed_5
    lasting 15 timesteps happened at the current timestep
    (T=4640ms), what would most likely happen next?
    Answer only with a 4 letter string using F and T ...",
  "options": {
    "A": "No significant effect: safety stays normal ...",
    "B": "Following the event, command and measured TCP ...",
    "C": "Following the event, the system remains ...",
    "D": "Following the event, robot current increases ..."
  },
  "answer": "FTTF",
  "provenance": {
    "dataset": "inter_aursad",
    "episode": "experiment_1",
    "subseries_start_index": 131,
    "subseries_length": 62
  },
  "context": {...}
}
```